



Data Article

ONUBAD: A comprehensive dataset for automated conversion of Bangla regional dialects into standard Bengali dialect



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ABSTRACT

Despite significant research on the Bangla language in Natural Language Processing (NLP), there remains a notable resource deficit for its diverse regional dialects, such as those spoken in Chittagong, Sylhet, and Barisal. These dialects, often considered unintelligible to speakers of Standard Bengali, pose challenges due to their unique grammatical structures and phonetic variations. Some linguists categorize them as distinct languages. To address this, we present ONUBAD, a large and freely available dataset for the automatic translation of Chittagong, Sylhet, and Barisal dialects into Standard Bangla using a Neural Machine Translation (NMT) system. ONUBAD provides a parallel corpus of 1540 words, 130 clauses, and 980 sentences per regional dialect and their standard counterparts along with English translation. The dataset includes metadata on phonetic variations and grammatical features, aiming to bridge the gap between standard and non-standard forms of Bangla. It serves as a valuable resource for researchers in NLP, dialect studies, and linguistic preservation, helping to develop more accurate and

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contextually relevant translation models. The dataset was collected between July and September 2024 from diverse sources such as books, websites, and regional people with the help of regional dialect specialists. It is hosted by the Department of Computer Science and Engineering, Jahangirnagar University, and is freely accessible at <https://data.mendeley.com/datasets/6ft99kf89b/2>

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Specifications Table

Subject	Computer Science/Artificial Intelligence
Specific subject area	Translation of Bangla Regional Dialects into Standard Bengali Dialect
Data Format	Text/String, Raw
Type of data	Table (.xlsx format)
Data collection	Data was extracted from various websites [1,2], and people (volunteers) residing in the districts of Chittagong, Sylhet, and Barisal in Bangladesh. Data from volunteer were collected with their informed consent, confirming that they understood the objectives of the study, the significance of their participation, and their right to discontinue participation at any moment. Furthermore, data obtained from the websites was done so with permission and in accordance with the conditions of use of the relevant platform. The dataset was selectively collected to ensure balanced representation across different data labels. The dataset encompasses three regional dialects: Chittagong, Sylhet, and Barisal along with English translation. Each dialect contains 2650 entries, comprising 1540 words, 130 clauses, and 980 sentences. Additionally, the data has been annotated by native experts in Bangla regional dialects.
Data source location	Jahangirnagar University, Savar, Dhaka-1342, Bangladesh.
Data accessibility	Repository name: Mendeley Data Data identification number: 10.17632/6ft99kf89b.2 Direct URL to data: https://data.mendeley.com/datasets/6ft99kf89b/2
Related research article	None

1. Value of the Data

- Translating regional dialects into Standard Bengali is typically done manually by linguists, which can be time-consuming and prone to inconsistencies. Developing a data-driven translation model is imperative to ensure consistency among texts and expedite translation operations.
- This dataset captures the most frequently regional words, clauses, and sentences. It offers greater diversity in regional dialects including Chittagong, Sylhet, and Barisal compared to other datasets [3], providing a more comprehensive resource for developing robust translation models and linguistic tools.
- This dataset captures regional linguistic features, their Bengali and English translations, aiding researchers in developing tools for automatic translation and linguistic analysis.
- The dataset can be used to train, test, and evaluate various natural language processing models, aiding in the development of tools for dialect translation, linguistic research, and improving accessibility to Chittagong, Sylhet, and Barisal-speaking populations through technology.
- Translating regional languages into Standard Bengali improves communication between local farmers and agricultural extension services, ensuring accurate dissemination of critical farming instructions, while also supporting the creation of multilingual resources and training materials that make advanced agricultural practices more accessible to Chittagong, Sylhet, and Barisal-speaking farmers.

- This dataset can help develop educational tools that bridge the gap between regional and Standard Bengali, fostering a more inclusive learning environment for Chittagong, Sylhet, and Barisal-speaking students.

2. Background

Bangla is the seventh most spoken language in the world. Approximately 272.7 million people in Bangladesh and some parts of India, as well as a significant diaspora worldwide, speak Bangla. People speak several different dialects of Bangla in different regions of Chittagong, Sylhet, and Barisal. It has been estimated that around 13 million people in Bangladesh speak Chittagonian dialect [4]. Approximately 11.8 million people in the Sylhet region of Bangladesh including Assam, northern Tripura, and Manipur in India speak the Sylheti dialect. Also, about 400,000 people speak Sylheti in the UK. Nearly 549,031 people in the Barisal Division of Bangladesh speak the Barisal dialect [5]. The native Bangla speakers find it difficult to understand these regional dialects. So, there arises a need to solve the communication problem for which an extensive dataset is required.

This dataset is compiled to support the development of sophisticated translation models and linguistic analysis tools, facilitating efficient translation between the regional dialects to standard Bangla. As the Bangla regional dialects have a varied range of vocabulary, pronunciation, and grammar, there is a need to address this linguistic variety. The dataset was collected from websites and volunteers, to identify these variances in dialects to facilitate more accurate translation to enhance the understanding of regional dialects within the broader Bangla linguistic landscape.

3. Data Description

This dataset is valuable for developing natural language processing tools, such as translation models and regional language understanding systems, and promotes linguistic diversity by enabling the digital representation of less-resourced regional languages in Bangladesh. The dataset consists of translations from Standard Bangla to three major regional languages: Chittagong, Sylheti, and Barisal dialects and corresponding English translation. It contains sentences in Standard Bangla, each translated into their respective regional variations, aiming to capture the linguistic and cultural nuances unique to each region. The dataset contains three files named Word, Clause, and Sentence. The file description will be discussed in this section. Table 1 presents the details of the ONUBAD dataset.

The dataset overview in numbers including all numerical details, such as total count, maximum, minimum, and average values of the dataset is illustrated in Table 2.

The data was collected from different websites, Literature, news articles, and regional people who reside in Chittagong, Sylhet, and Barisal regions. We collaborated with a team of skilled and qualified translators, each possessing specialized knowledge of their respective regions, to develop the "ONUBAD" dataset. Their expertise and linguistic proficiency were crucial in ensuring the dataset's accuracy and reliability.

Table 1
Details of the ONUBAD dataset.

Category	Word	Clause	Sentence	Total
Chittagong	1540	130	980	2650
Sylhet	1540	130	980	2650
Barisal	1540	130	980	2650
Standard Bangla	1540	130	980	2650
English Translation	1540	130	980	2650

Table 2
Dataset Overview in Numbers (This section includes all numerical details, such as maximum, minimum, and average word count values of our dataset).

	Minimum Word Count (Clause)	Maximum Word Count (Clause)	Average Word Count (Clause)	Minimum Word Count (Sentence)	Maximum Word Count (Sentence)	Average Word Count (Sentence)
Chittagong	1	4	2	1	11	3
Sylhet	1	4	2	1	10	3
Barisal	1	4	2	1	10	3
Standard Bangla	1	4	2	2	10	3
English Translation	1	8	2	2	14	3

Table 3
Examples of the dataset.

Categories	Regional Dialect				
	Chittagong Language	Sylhet Language	Barisal Language	Standard Bangla Language	English Translation
Word	আই	আমি	মুই	আমি	I
Clause	আঁরার জুরান-আন	মোরারে ভাষাটা	মোগো ভাষাটা	আমাদের ভাষাটা	Our language
Sentence	আঁই কিরকেট খেইল্যম	আমি ক্রিকেট খেলমু	মুই ক্রিকেট খেলমু	আমি ক্রিকেট খেলব	I will play cricket

The ONUBAD dataset is used to train models for automatic translation of Bangla regional dialects into standard Bengali language. The dataset contains 2650 observations, each containing a word, clause, and sentence. Table 3 shows examples of the dataset we employed.

4. Experimental Design, Materials and Methods

After the development of the dataset, we meticulously ensured that it could be utilized for multi-class classification problems, particularly in recognizing Sylhet, Chittagong, and Barisal dialects. Therefore, to resolve a classification problem using our dataset, the steps mentioned in Fig. 1 should be conducted to demonstrate its usability.

Data pre-processing is essential before using the dataset in downstream tasks to enhance the accuracy of the classifier. This involves removing unnecessary punctuation, special characters, and Bangla stop words from the sentences that hold no significance in deep learning model

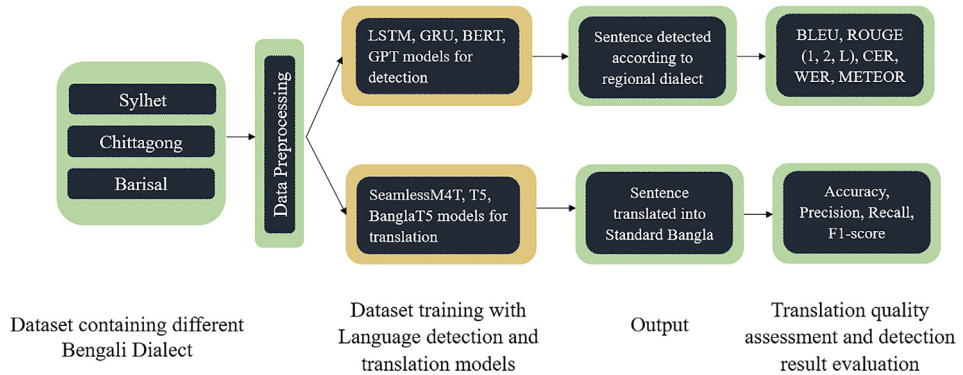


Fig. 1. The proposed workflow diagram of the dataset evaluation experiment.

training. Tokenization is a method that divides the input sentences into individual words or tokens, which serve as the fundamental units for subsequent processing. Additionally, employing lemmatization and stemming techniques helps identify word roots, and grouping various forms of a word can significantly enhance the experimentation results. These tokens and base words obtained from lemmatization are then forwarded to the following stages of the seq2seq models and transformer-based models to enable efficient language processing and translation.

Various seq2seq models and transformer-based models such as LSTM, GRU, BERT, and GPT can be employed for detection, and SeamlessM4T, T5, and BanglaT5 models can be used for regional dialect translation into standard Bangla. To determine the quality of the translations from regional dialects to standard Bangla widely recognized metrics can be utilized in the generation task, such as CER, WER, METEOR, BLEU, and ROUGE (ROUGE-1, ROUGE-2, and ROUGE-L) scores [6,7]. This analysis compares the performance of individual translation models and gives qualitative insights into translation quality. Evaluation metrics such as accuracy, precision, recall, and F1-score can be utilized to assess the efficiency of the language-translation models. These metrics provide valuable insights into the quality and accuracy of the machine translations produced by the models. The proposed workflow diagram of the dataset evaluation experiment is illustrated in Fig 1.

Limitations

Not applicable.

Ethics Statement

We have read and followed the ethical requirements for publication in Data in Brief. We confirm that the current work does not involve animal experiments. We ensure that all of the data utilized in this work was obtained from the appropriate primary sources and reused with their consent. The authors have complied with ethical standards and, when necessary, have acquired permission to reuse the data. The privacy and confidentiality of all the volunteers were protected throughout the study. The datasets used in the article are open to the public. For the usage of these datasets, proper citation rules should be maintained.

Credit Author Statement

Nusrat Sultana: Conceptualization, Writing - Original Draft, Methodology, Data curation, Software. **Rumana Yasmin:** Writing - Original Draft, Methodology, Investigation, Software. **Bijon Mallik:** Methodology, Software. **Mohammad Shorif Uddin:** Supervision, Writing - Review & Editing.

Data Availability

ONUBAD: An Extensive Dataset for Automated Translation of Bangla Regional Dialects into Standard Bangla Language (Original data) (Mendeley Data).

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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